HLDS Screen Relay

AMXX Secure Two-Way Conduit Architecture Documentation

A high-performance, two-way terminal console relay designed to manage Half-Life Dedicated Servers (HLDS) and AMX Mod X (AMXX) engine sessions securely. This utility completely bypasses the ancient, insecure plaintext UDP RCON protocol by establishing an encrypted, multiplexed control plane directly through OpenSSH into a remote GNU screen instance.

Key System Features

- **Encrypted Transport Layer:** Uses asymmetric id_ed25519 key pairs to secure actions.
- **OpenSSH Multiplexing:** Spawns a persistent local background socket pool for instant (0ms) updates.
- **Sliding View Buffer:** Type Spacebar then Enter to dynamically grow your history buffer upward.
- **Live Peekaboo Relay Mode:** Type 'relay' to shift into a scrolling, real-time telemetry screen. Ctrl+C drops back safely.
- **8-Bit Aesthetic:** Moving ANSI color-raster lines, white-on-blue environment, and retro 440Hz alert tones.

```
**** COMMODORE 64 BASIC V2 ****
64K RAM SYSTEM 38911 BASIC BYTES FREE

READY.
```

Compilation Commands

Linux Native:

```
g++ -O2 amxx_remote_client.cpp -o amxx_launcher -lreadline
```

Windows Cross-Build:

```
i686-w64-mingw32-g++ -O3 amxx_remote_client.cpp -o amxx_remote_32.exe -
```

Windows 0ms Performance Optimization

Add this to your C:\Users\<User>\.ssh\config file:

```
Host *
    ControlMaster auto
    ControlPath ~\AppData\Local\Temp\ssh-%r@%h:%p
    ControlPersist 10m
```